Seminar 8: The z-Transform

Topics: definition and properties of the z-transform, system functions, time shifting and unit delays, convolution and polynomial multiplication, cascaded systems, factorisation and inverse filtering, the unit circle, poles and zeros, nulling filters, running-sum and band-pass filters, and linear-phase FIR systems.

Key idea. The z-transform converts time-domain operations into algebra: delays become powers of z^{-1}, convolution becomes multiplication, and poles and zeros reveal important properties of the filter and its frequency response.

1. From a sequence to its z-transform

Consider the finite-length sequence $x[n] = \{2, -1, 3, 4\}, \;\; 0 \le n \le 3$

(a) Write the z-transform $X(z)$ as a polynomial in z^{-1}.

(b) State the coefficient of each power $z^{0}, z^{-1}, z^{-2}, z^{-3}$ and explain how it corresponds to the samples of $x[n]$.

(c) Given $Y(z) = 5 - 2z^{-1} + z^{-3}$, recover the corresponding finite sequence $y[n]$.

(d) Write the z-transform of the shifted impulse $\delta[n-5]$.

(e) Explain why, for a finite sequence, the z-transform may be viewed as a polynomial whose variable is z^{-1}.

2. Time shifting and the unit-delay operator

Let $x[n] = \{3, 1, 4, 1, 5\}$

(a) Write $X(z)$.

(b) Define $y[n] = x[n-2]$. Use the time-shifting property to obtain $Y(z)$ without re-summing the sequence.

(c) Explain why multiplication by z^{-1} represents a one-sample delay.

(d) What time-domain operation is represented by multiplication by z^{-4}?

(e) Draw a block diagram containing the required unit delays to generate $x[n-3]$ from $x[n]$.

3. System function of an FIR filter

An FIR system is described by $y[n] = 2x[n] - 3x[n-1] + x[n-2]$

(a) Determine the impulse response $h[n]$.

(b) Determine the system function $H(z)$.

(c) Show explicitly that $H(z)$ is the z-transform of $h[n]$.

(d) Obtain the frequency response by evaluating the system function on the unit circle, $z = e^{\{j\hat{\omega}\}}$.

(e) Evaluate $H(e^{\{j\hat{\omega}\}})$ at $\hat{\omega} = 0$ and $\hat{\omega} = \pi$ and interpret the results.

4. Convolution as polynomial multiplication

Let $x[n] = \{1, 2, 1\}, \quad h[n] = \{1, -1, 2\}$

(a) Write $X(z)$ and $H(z)$.

(b) Multiply the two z-domain polynomials to obtain $Y(z) = X(z)H(z)$.

(c) Read the coefficients of $Y(z)$ to obtain the output sequence $y[n]$.

(d) Verify the same result by direct convolution in the time domain.

(e) Explain why polynomial multiplication and convolution are two descriptions of the same operation.

5. Cascaded systems and factorisation

Two FIR filters have system functions $H_1(z) = 1 - z^{-1}, \quad H_2(z) = 1 + 2z^{-1} + z^{-2}$

(a) Determine the overall system function $H(z)$ when the filters are cascaded.

(b) Write the equivalent single difference equation relating $x[n]$ and $y[n]$.

(c) Find the equivalent impulse response using convolution.

(d) Show that reversing the order of the two filters gives the same overall system.

(e) Suppose instead that $H(z) = 1 + z^{-1} - z^{-2} - z^{-3}$. Factor it into lower-order FIR sections.

6. Inverse filtering and deconvolution

A system has $H_1(z) = 1 - 0.5z^{-1}$

(a) Find the formal inverse system function $H_2(z)$ such that $H_1(z)H_2(z) = 1$.

(b) Explain why applying $H_2(z)$ after $H_1(z)$ is called deconvolution or inverse filtering.

(c) Write the first four terms of the power-series expansion of $H_2(z)$ in powers of z^{-1}.

(d) What does this expansion suggest about the duration of the inverse filter impulse response?

(e) Explain why an exact inverse of a finite FIR filter need not itself be FIR.

7. Poles and zeros in the z-plane

Consider $H(z) = 1 - 1.5z^{-1} + 0.5z^{-2}$

(a) Multiply by an appropriate power of z and factor the numerator polynomial.

(b) Determine the zeros of $H(z)$.

(c) Identify the poles introduced by the negative powers of z.

(d) Plot the poles and zeros in the complex z-plane and include the unit circle.

(e) Which zero lies on the unit circle? What normalized frequency does it correspond to?

8. The unit circle and frequency response

For the system $H(z) = (1 - e^{j\pi/3}z^{-1})(1 - e^{-j\pi/3}z^{-1})$

(a) Expand $H(z)$ to obtain real FIR coefficients.

(b) Plot the two zeros in the z-plane.

(c) Evaluate the system function on the unit circle to obtain $H(e^{j\hat{\omega}})$.

(d) Without performing a full magnitude derivation, identify the frequencies for which $|H(e^{j\hat{\omega}})| = 0$.

(e) Explain geometrically why a zero on the unit circle causes zero gain at the angle of that zero.

9. Designing a real sinusoid-nulling filter

Design a second-order real FIR filter that completely suppresses $x[n] = \cos(\pi n/4)$

(a) Determine the two unit-circle zero locations required to suppress the positive- and negative-frequency complex exponentials.

(b) Write the factored system function $H(z)$.

(c) Expand the factors and obtain the real filter coefficients.

(d) Write the corresponding difference equation.

(e) Verify directly that $H(e^{j\pi/4}) = 0$.

(f) Would this filter also remove $\cos(3\pi n/4)$ exactly? Explain from the zero locations.

10. Running-sum filter and roots of unity

Consider the length-6 running-sum filter $H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5}$

(a) Use the finite geometric-series formula to rewrite $H(z)$ in rational form.

(b) Determine all six roots of $z^6 = 1$ and explain why the root at $z = 1$ is cancelled in $H(z)$.

(c) Plot the remaining zeros and the poles at the origin.

(d) List all normalized frequencies in $-\pi < \hat{\omega} \leq \pi$ that are nulled by the filter.

(e) Determine the DC gain and explain why the running-sum filter has a large response near DC.

11. From a running-sum filter to a complex band-pass filter

A length-8 complex band-pass filter is formed by omitting one root of unity from a running-sum structure. Let the omitted root correspond to $k_0 = 2, \quad \hat{\omega}_0 = 2\pi k_0/8 = \pi/2$

(a) List the eight roots of unity and identify the omitted root.

(b) Explain why the response is largest near $\hat{\omega} = \pi/2$ rather than near DC.

(c) Sketch the pole-zero pattern in the z-plane.

(d) State which normalized frequencies are exactly nulled.

(e) Explain why the coefficients of this shifted filter are generally complex.

12. Real-coefficient band-pass filter

A real band-pass FIR filter can be constructed by combining conjugate complex band-pass filters centred at $\hat{\omega} = \pm\pi/3$

(a) Explain why both passbands are required if the filter coefficients are real.

(b) Describe the conjugate-symmetry condition on the coefficients or frequency response that guarantees a real output for a real input.

(c) Sketch a qualitative magnitude response over $-\pi \leq \hat{\omega} \leq \pi$ showing the two passbands.

(d) Sketch a corresponding qualitative zero pattern on the unit circle.

(e) Compare the real band-pass response with a single complex band-pass response centred only at $+\pi/3$.

13. Symmetry and linear phase

Consider the symmetric FIR impulse response $h[n] = \{1,2,3,2,1\}$

(a) Write the system function $H(z)$.

(b) Factor out the centre delay and group symmetric terms.

(c) Evaluate on the unit circle and use $e^\{j\theta\} + e^\{-j\theta\} = 2\cos\theta$ to express the result as a real amplitude factor times a linear-phase factor.

(d) Determine the phase delay in samples.

(e) Explain why symmetry of the FIR coefficients leads to linear phase.

14. Integrated z-domain analysis of an FIR system

Consider $H(z) = (1 - z^{-1})(1 - 2\cos(\pi/3)z^{-1} + z^{-2})$

(a) Expand $H(z)$ and write the equivalent FIR difference equation.

(b) Determine and plot all zeros and poles.

(c) Identify the normalized frequencies completely suppressed by the filter.

(d) For input $x[n] = 1 + \cos(\pi n/3) + \cos(\pi n/2)$, identify which components are removed and which can remain in the steady-state output.

(e) Explain how the same conclusions can be reached from the factorisation without explicitly convolving the input with the impulse response.

15. Computational investigation with Python

Use NumPy, Matplotlib, and scipy.signal to connect the time, z, and frequency domains.

(a) For $h[n] = \{1, -1,2,1\}$, form $H(z)$ and use $\mathrm{numpy.roots}$ to determine its zeros.

(b) Plot the zeros and poles in the z-plane together with the unit circle.

(c) Use scipy. signal. freqz to plot $|H(e^{\{j\hat{\omega}\}})|$ and phase; relate deep minima to nearby unit-circle zeros.

(d) Choose a short input sequence, compute the output both with numpy. convolve and by multiplying the z-polynomials, and verify that the results agree.

(e) Factor a higher-order FIR polynomial into lower-order sections and verify numerically that cascading the sections gives the same output as the single filter.

(f) Design a second-order nulling filter for $\hat{\omega}_0 = 0.4\pi$ and verify from both the zero plot and freqz that the response vanishes at that frequency.